

The lowest energy conformations are the two chair conformations, and therefore, cyclohexane will spend the majority of its time in a chair conformation. Accordingly, the remainder of our treatment of cyclohexane will focus on chair conformations. Our first step is to master drawing them.

4.11 Drawing Chair Conformations

When drawing a chair conformation, it is important to draw it precisely. Make sure that you avoid drawing sloppy chairs, because it will be difficult to draw the substituents correctly if the skeleton is not precise.

Drawing the Skeleton of a Chair Conformation

Let's get some practice drawing chairs.

SKILLBUILDER



4.9 DRAWING A CHAIR CONFORMATION

LEARN the skill

Draw a chair conformation of cyclohexane:



SOLUTION

The following procedure outlines a step-by-step method for drawing the skeleton of a chair conformation precisely:

- | | | |
|--------|--|--|
| Step 1 | | Draw a wide V . |
| Step 2 | | Draw a line going down at a 60° angle, ending just before the center of the V. |
| Step 3 | | Draw a line parallel to the left side of the V, ending just before the left side of the V. |
| Step 4 | | Draw a line parallel to the line from Step 2, going down exactly as low as that line. |
| Step 5 | | Connect the dots. |

When you are finished drawing a chair, it should contain three sets of parallel lines. If your chair does not contain three sets of parallel lines, then it has been drawn incorrectly.



PRACTICE the skill

4.21 Practice drawing a chair several times using a blank piece of paper. Repeat the procedure until you can do it without looking at the instructions above. For each of your chairs, make sure that it contains three sets of parallel lines.

APPLY the skill

4.22 Chemists have developed micron-sized, doughnut-shaped structures that might be used someday to deliver drugs across cell membranes.⁹ During a procedure to prepare such micron-sized structures, dioxane (shown below) was found to be a crucial co-solvent, ensuring a good yield of structures with the desired shape and size. Draw a chair conformation of dioxane.



Dioxane

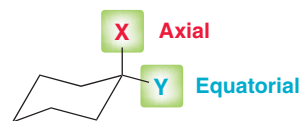


Drawing Axial and Equatorial Substituents

Each carbon atom in a cyclohexane ring can bear two substituents (Figure 4.21). One group is said to occupy an **axial position**, which is parallel to a vertical axis passing through the center of the ring. The other group is said to occupy an **equatorial position**, which is positioned approximately along the equator of the ring (Figure 4.21). In order to draw a substituted cyclohexane, we must first practice drawing all axial and equatorial positions properly.

FIGURE 4.21

Axial and equatorial positions in a chair conformation.



SKILLBUILDER



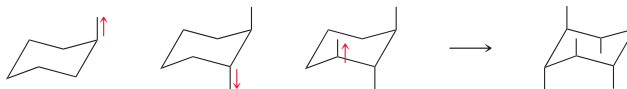
4.10 DRAWING AXIAL AND EQUATORIAL POSITIONS

LEARN the skill

Draw all axial and all equatorial positions on a chair conformation of cyclohexane.

SOLUTION

Let's begin with the axial positions, as they are easier to draw. Begin at the right side of the V and draw a vertical line pointing up. Then, go around the ring, drawing vertical lines, alternating in direction (up, down, up, etc.):

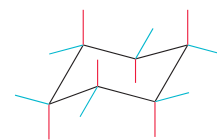


These are the six axial positions. All six lines are vertical.

Now let's draw the six equatorial positions. The equatorial positions are more difficult to draw properly, but mistakes can be avoided in the following way. We saw earlier that a properly drawn chair skeleton is composed of three pairs of parallel lines. Now we will use these pairs of parallel lines to draw the equatorial positions. In between each pair of red lines, we draw two equatorial groups that are parallel to (but not directly touching) the red lines:



Notice that all equatorial positions are drawn going to the outside of, or away from, the ring, not going into the ring. Now let's summarize by drawing all six axial positions and all six equatorial positions.



STEP 1
Draw all axial positions as vertical lines alternating in direction.

STEP 2
Draw all equatorial positions as pairs of parallel lines.

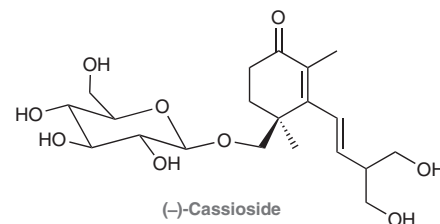
PRACTICE the skill

4.23 Practice drawing a chair conformation with all six axial positions and all six equatorial positions. Practice several times on a blank piece of paper. Repeat until you can draw all twelve positions without looking at the instructions above.

APPLY the skill

4.24 (–)-Cassioside, an anti-ulcer agent isolated from the dried stem bark of Chinese cinnamon,¹⁰ contains two six-membered rings, one of which adopts a chair conformation as shown.

- Identify each group attached to the chair as being in an axial or equatorial position.
- Draw each of the missing hydrogen atoms connected to this six-membered ring, and properly position each in an axial or equatorial position, as appropriate.

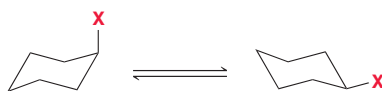


need more PRACTICE? Try Problems 4.46, 4.59

4.12 Monosubstituted Cyclohexane

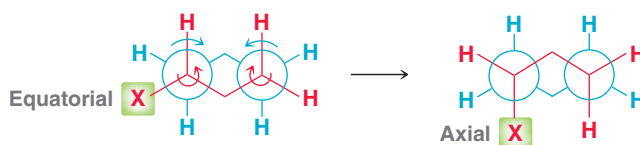
Drawing Both Chair Conformations

Consider a ring containing only one substituent. Two possible chair conformations can be drawn: The substituent can be in an axial position or in an equatorial position. These two possibilities represent two different conformations that are in equilibrium with each other:



The term “ring flip” is used to describe the conversion of one chair conformation into the other. This process is not accomplished by simply flipping the molecule like a pancake. Rather, a **ring flip** is a conformational change that is accomplished only through a rotation of all C—C single bonds. This can be seen with a Newman projection (Figure 4.22). Let’s get some practice drawing ring flips.

FIGURE 4.22
A ring flip drawn with
Newman projections.



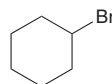
SKILLBUILDER



4.11 DRAWING BOTH CHAIR CONFORMATIONS OF A MONOSUBSTITUTED CYCLOHEXANE

LEARN the skill

Draw both chair conformations of bromocyclohexane:



STEP 1

Draw a chair conformation.

STEP 2

Place the substituent.

SOLUTION

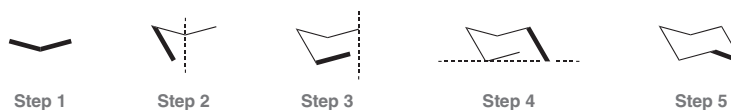
Begin by drawing the first chair conformation, and place the bromine in any position:



In this chair conformation, the bromine occupies an axial position. In order to draw the other chair conformation (the result of a ring flip), redraw the skeleton of the chair. Only this time, we will draw the skeleton differently. The first chair was drawn by following these steps:



The second chair must now be drawn by *mirroring* these steps:



STEP 3

Draw a ring flip, and the axial group should become equatorial.

In the second chair, the bromine will occupy an equatorial position:

